Chirp Based Waveform Design for Next Generation Non Terrestrial Networks

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Outline

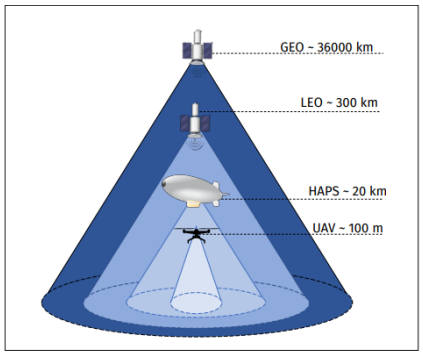
1. Context
2. Spectral Analysis of the AFDM Modulation
3. LoRa-Based Chirp Division Multiplexing
4. Future Works

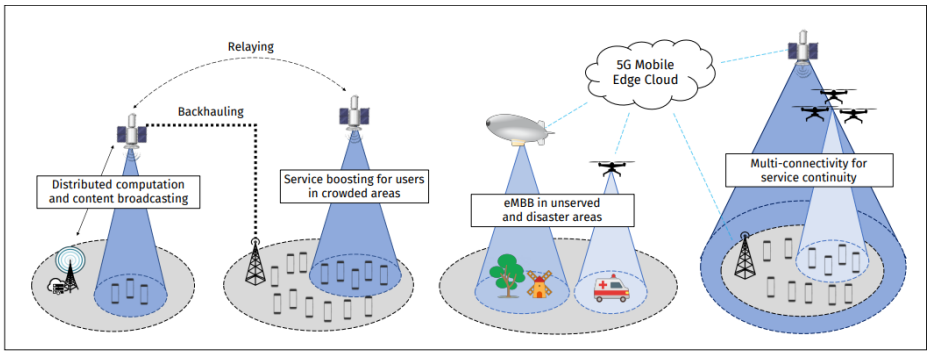
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Broadband Communications (6G)

- Next-generation networks (6G) demand **higher data rates**, **global 3D connectivity**, **integrated sensing and communication (ISAC)**, and a **unified architecture**.
- Non-Terrestrial Networks (NTN) meet these demands but introduce many challenges, primarily **high mobility**. [1]



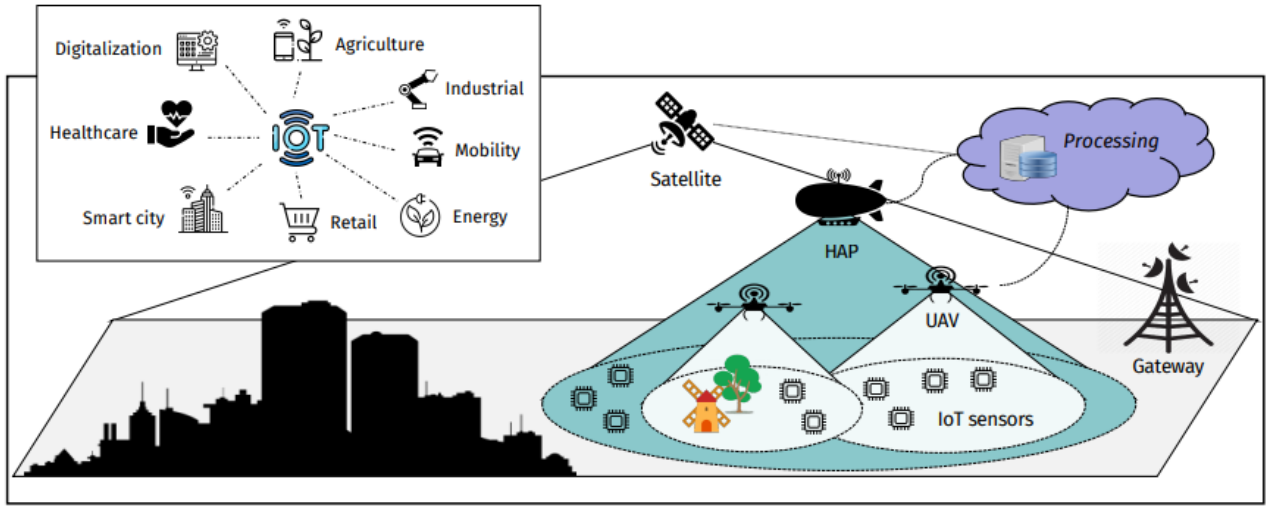


- **The Need for Resilient Waveforms** to ensure high spectral efficiency, robust communication links, and reliable sensing under severe NTN channel conditions.

[1] Image credit: Giordani et al *Non-terrestrial networks in the 6G era: Challenges and opportunities*, IEEE Network.

Satellite IoT (DtS IoT)

- Global IoT scaling drives the demand for **ubiquitous coverage** and **higher data rates** without sacrificing long range capabilities.
- NTN enables Direct-to-Satellite (DtS) IoT, significantly extending network reach and capacity. [2]



- **Waveform Design:** Developing solutions able to support massive IoT expansion and dense, simultaneous device transmissions.

[2] Image credit: Wang et al *On the performance of non-terrestrial networks to support the internet of things*, Asilomar.

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Motivation

- Beyond 5G and 6G systems require the support of high mobility applications.
- The AFDM alternative: A chirp-based multicarrier waveform.
- Although performance gains in power efficiency are well studied, the spectral characteristics of AFDM remain under-explored. In this work we [3]:
  - Provide a rigorous analytical derivation of the continuous-time AFDM signal ...
  - and its power spectral density (PSD).
  - Study the influence of different parameters, notably the chirp rate, on out-of-band (OOB) emissions.

[3] El Marzougui, Omar and Thomas, Nathalie and Poulliat, Charly, *Spectral Analysis Of The AFDM Modulation*, 34th European Signal Processing Conference (EUSIPCO 2026).

Signal Model

- AFDM transmits symbols using a bank of chirps performing multiple frequency sweeps over the interval $[0, B]$. It is characterized by an integer C, representing the number of full frequency sweeps.
- For a set of N chirps over a duration $T = N/B$, the instantaneous frequency of the k-th chirp is given by the following formula, with $0 \leq t \leq T$, $0 \leq k \leq N-1$ and $1 \leq l \leq C$:

$$f_k(t) = \frac{B}{N}k + 2c_1B^2t - B\sum_{l=1}^{C}\mathcal{H}(t-\tau_{l,k}), \quad c_1 = \frac{C}{2N}, \quad \tau_{l,k} = \frac{Nl-k}{2Nc_1B}$$

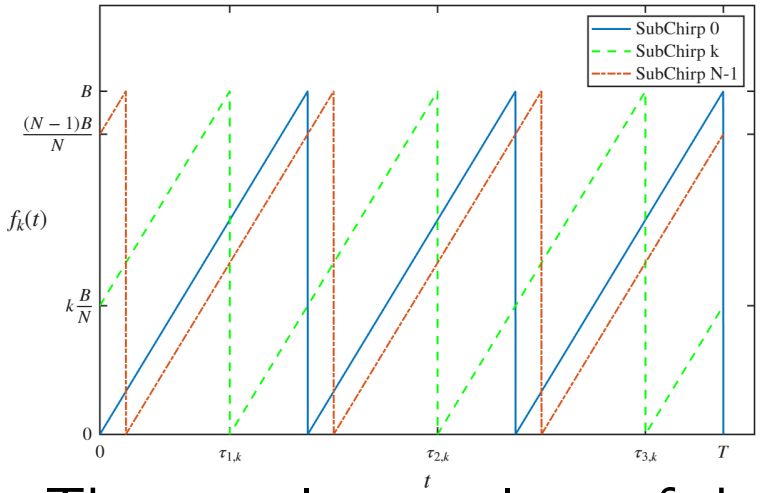
- The chirp rate is adjusted to the Doppler such that: $c_1 = \frac{2\lceil f_{max}T\rceil+1}{2N}$.
- $\tau_{l,k}$ are the frequency reset times to 0, temporally spaced by $T_C = \tau_{l+1,k} - \tau_{l,k} = \frac{T}{C}$.

Signal Model

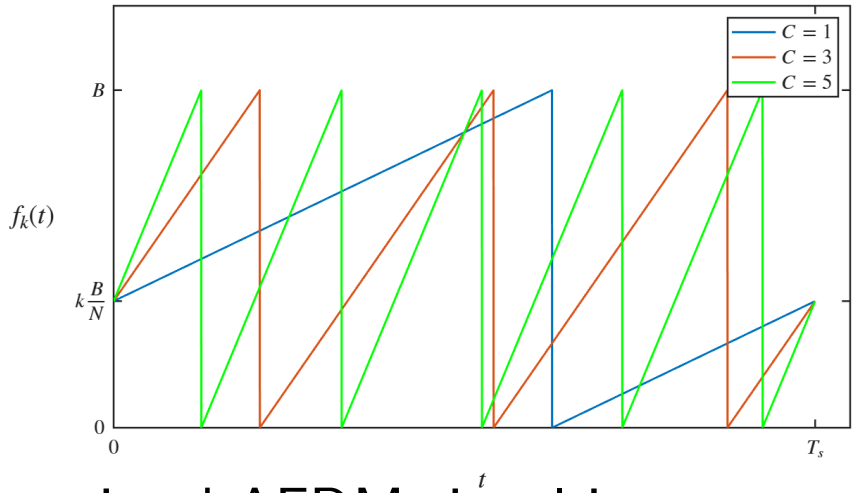
- The phase of the AFDM chirp is expressed as follows:

$$\phi_k(t) = 2\pi\left[\frac{B}{N}kt + c_1 B^2 t^2 - B\sum_{l=1}^{C}(t-\tau_{l,k})\mathcal{H}(t-\tau_{l,k}) + c_2 k^2\right]$$

AFDM Chirp Bank



Different Chirp Rates



- The complex envelope of the transmitted AFDM signal is:

$$x(t) = \sum_n s(t-nT)g_T(t-nT)$$

where $s(t)$ is the AFDM symbol: $s(t) = \sum_{k=0}^{N-1} s_k e^{j\phi_k(t)}$.

Analytical PSD Formula

- Assuming that the transmitted data symbols are statistically independent and identically distributed (i.i.d.) with zero mean, the cross-correlation terms cancel out. Thus, the PSD of the AFDM signal is defined by the sum of the power spectra of its individual sub-chirps:

$$P(f) = \frac{\sigma_s^2}{T} \sum_{k=0}^{N-1} |\Phi_k(f)|^2$$

  Where $\Phi_k(f) = \int_0^T e^{j\phi_k(t)} e^{-j2\pi ft}\, dt$ is the Fourier transform of the sub-chirp $e^{j\phi_k(t)}$.

- We express the Fourier transform integral of the k-th sub-chirp for $0 \leq k \leq N-1$ and $1 \leq l \leq C$ as follows:

$$\Phi_k(f) = \int_0^T e^{j2\pi\left(\frac{B}{N}kt + c_1 B^2 t^2 - B\sum_{l=1}^{C}(t-\tau_{l,k})\mathcal{H}(t-\tau_{l,k}) + c_2 k^2 - ft\right)} dt$$

Analytical PSD Formula

- The idea is to divide the integral over the intervals $[\tau_{l,k}, \tau_{l+1,k}]$, which will result in a sum of integrals, and to express each term of this sum using Fresnel integrals:

$$C(x) = \int_0^x \cos(u^2\pi/2)du, \quad S(x) = \int_0^x \sin(u^2\pi/2)du$$

- Let $Z(a, b, t_1, t_2)$ be the function defined by $Z(a, b, t_1, t_2) = \int_{t_1}^{t_2} \exp\left[j2\pi(at + bt^2)\right] dt$. This function can be expressed in terms of Fresnel functions:

$$Z(a; b; t_1; t_2) = \frac{1}{2\sqrt{b}} e^{-j2\pi\frac{a^2}{4b}} \left[F\left(2\sqrt{b}\left(t_2 + \frac{a}{2b}\right)\right) - F\left(2\sqrt{b}\left(t_1 + \frac{a}{2b}\right)\right)\right]$$

- Where $F(x) = C(x) + jS(x)$.

Analytical PSD Formula

- Finally, using these tools, we can deduce an analytical expression of the Fourier transform of an AFDM sub-chirp using the $Z(.)$ function as follows:

$$\Phi_k(f) = \sum_{l=0}^{C} A_{k,l} \cdot Z\left(B\left(\frac{k}{N} - l\right) - f, c_1 B^2, \tau_{l,k}, \tau_{l+1,k}\right)$$

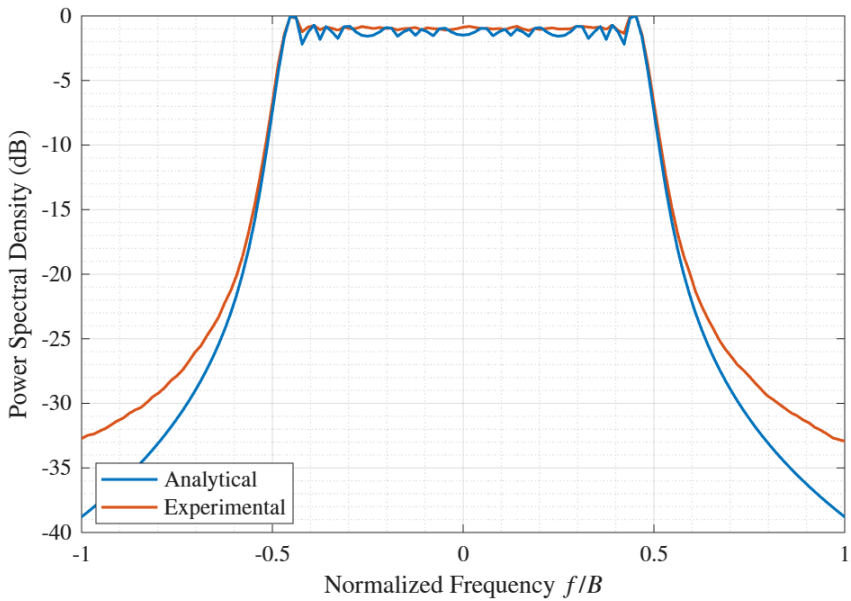
- Where $A_{k,l}$ is a phase coefficient independent of time t and which can be expressed as follows:

$$A_{k,l} = \exp\left[j2\pi\left(\frac{Nl(l+1)}{2C} - \frac{kl}{C} + c_2 k^2\right)\right]$$

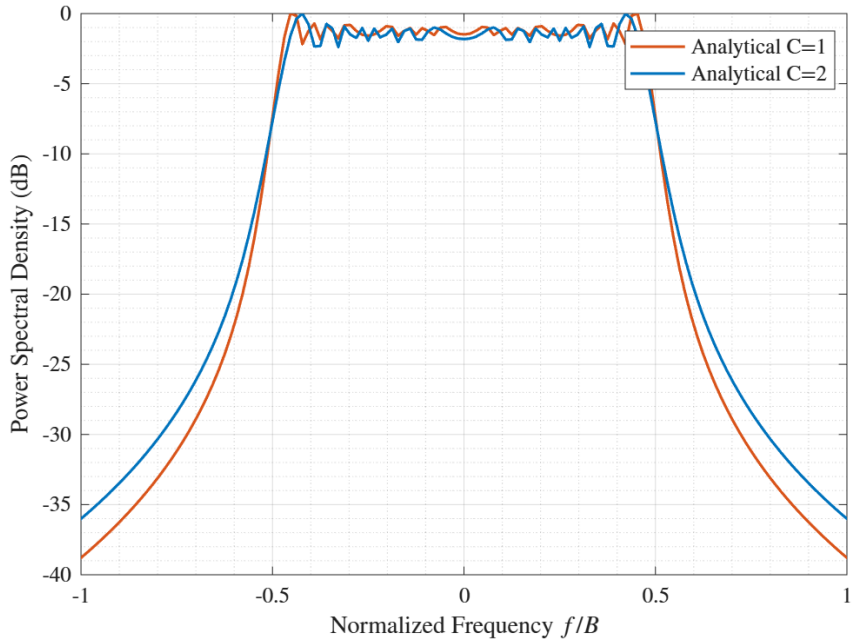
Numerical Results

- **Impact of the chirp rate:** A higher chirp rate leads to more significant out-of-band (OOB) emissions. This highlights a trade-off between Doppler robustness and spectral containment.

Analytical vs Experimental.



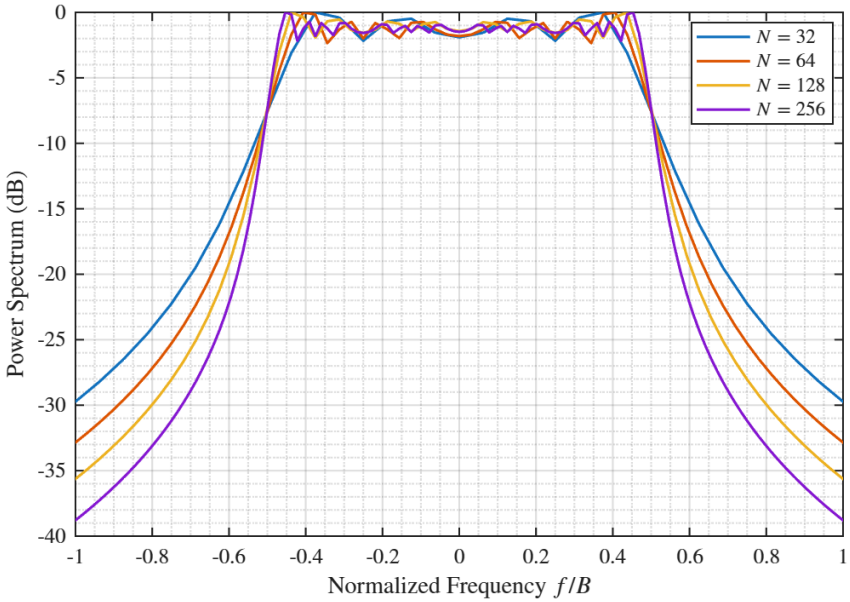
Influence of the Chirp Rate.



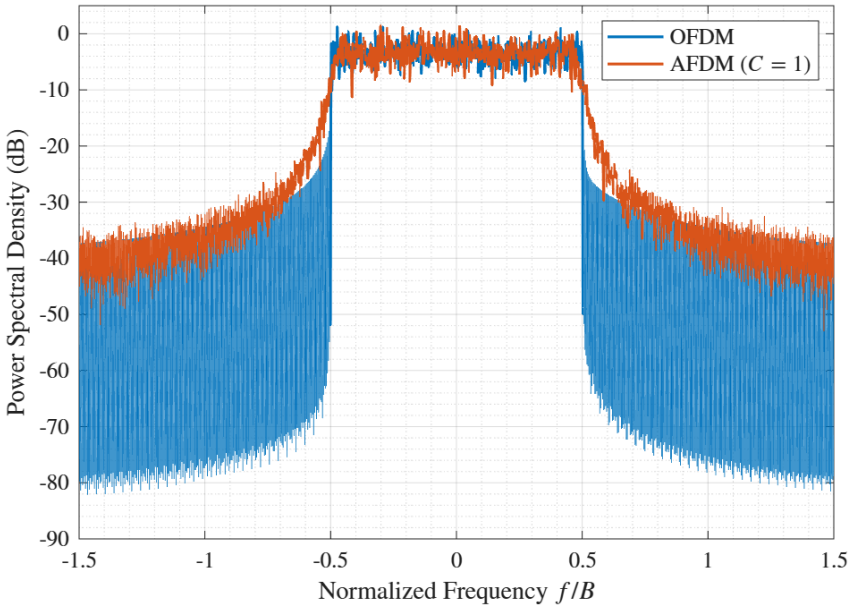
Numerical Results

- **Impact of sub-chirps (N):** Increasing the number of sub-chirps leads to a more compact spectrum, which translates into lower OOB emissions.
- **AFDM vs OFDM:** When using a rectangular window, the AFDM spectrum exhibits more OOB emissions compared to OFDM.

Influence of the N° of Sub-chirps.



AFDM vs OFDM.



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Motivation

- While LoRa CSS offers good long range performance and robust noise resilience, its spectral efficiency is remains limited because it spreads a single symbol over a wide bandwidth.
- As LPWANs scale, current extensions (e.g., IQ-LoRa, PSK-LoRa) provide only marginal capacity improvements and fall short of multicarrier division multiplexing schemes like NB-IoT.
- We develop a high data rate extension of LoRa CSS (named LoCDM) that [4]:
  - Overcomes LoRa's capacity limitations.
  - Achieves physical layer interoperability with NB-IoT through a shared digital architecture.

[4] El Marzougui, Omar and Thomas, Nathalie and Poulliat, Charly, *LoCDM: A New Chirp Division Multiplexing Waveform Using LoRa-Based Chirp Bank*, 2026 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit).

Signal Model

- The continuous-time LoRa CSS signal $0 \leq t \leq T$:

$$x_{lora}(t;s) = e^{j2\pi Bt\left(\frac{s}{N}-\frac{1}{2}+\frac{Bt}{2N}-\mathcal{H}\left(t-\frac{N-s}{B}\right)\right)}$$

- With N: the number of sub-chirps, B: the bandwidth, T: the chirp duration.
- Reuse of the chirp bank in division multiplexing mode:

$$x_{locdm}(t) = \sum_{k=0}^{N-1} s_k \phi_k(t), \quad \text{with } \phi_k(t) = e^{j2\pi Bt\left(\frac{k}{N}-\frac{1}{2}+\frac{Bt}{2N}-\mathcal{H}\left[t-\frac{N-k}{B}\right]\right)}$$

- Sampling at the chip rate $T_c = T/N$. Digital Implementation: $n \in \{0, 1, 2, \cdots, N-1\}$

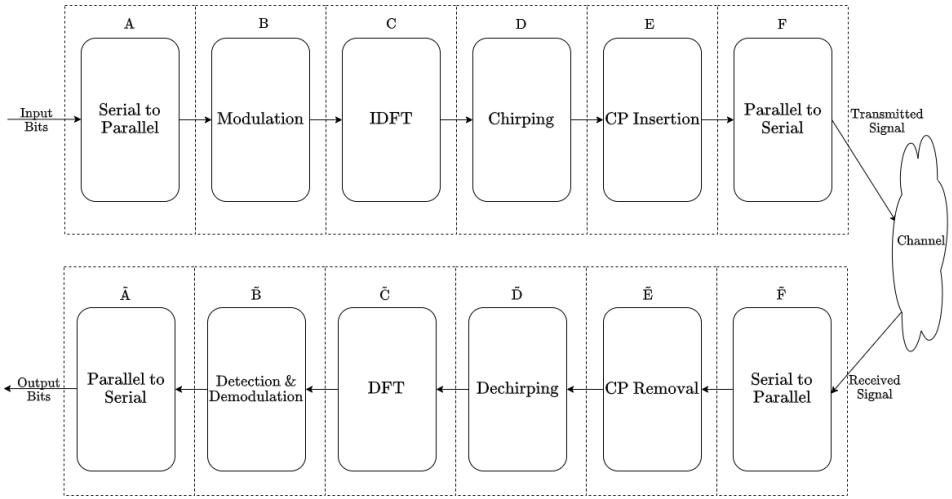
$$x_{locdm}(n) \triangleq \frac{1}{\sqrt{N}} x(t = nT_c) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} s_k e^{j2\pi n\left(\frac{k}{N}-\frac{1}{2}+\frac{n}{2N}\right)}$$

LoCDM as "Chirped NB-IoT"

- LoCDM modulation can be seen as a generalization of LoRa CSS aimed at improving spectral efficiency by simultaneously exploiting multiple frequency resources.
- Implementation via IDFT + Digital Chirping:

$$x_{locdm}(n) = \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} s_l e^{j2\pi \frac{nl}{N}} e^{j2\pi n\left(\frac{n}{2N}-\frac{1}{2}\right)} = \frac{1}{\sqrt{N}} \left(\sum_{l=0}^{N-1} s_l e^{j2\pi \frac{nl}{N}} \right) e^{\frac{j\pi}{N}(n^2-nN)}$$

- Physical (PHY) layer interoperability between LoRa CSS and NB-IoT:

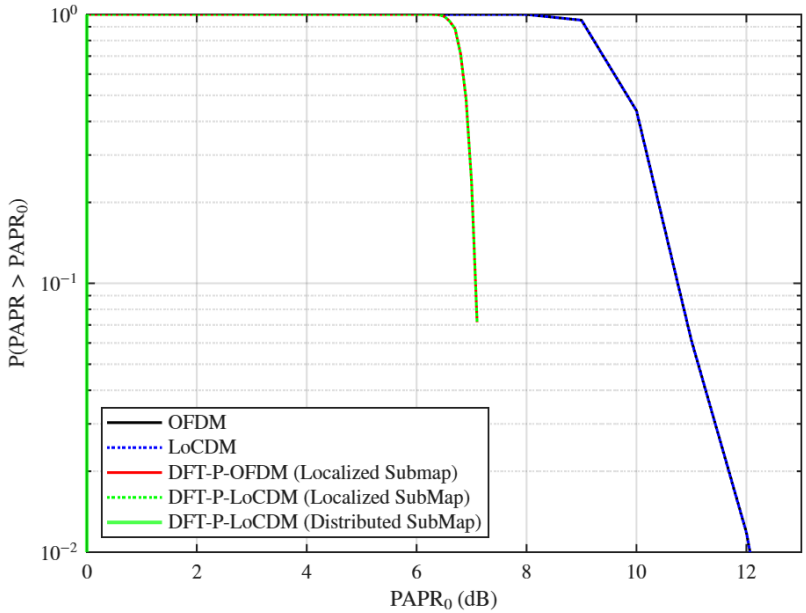


Performance

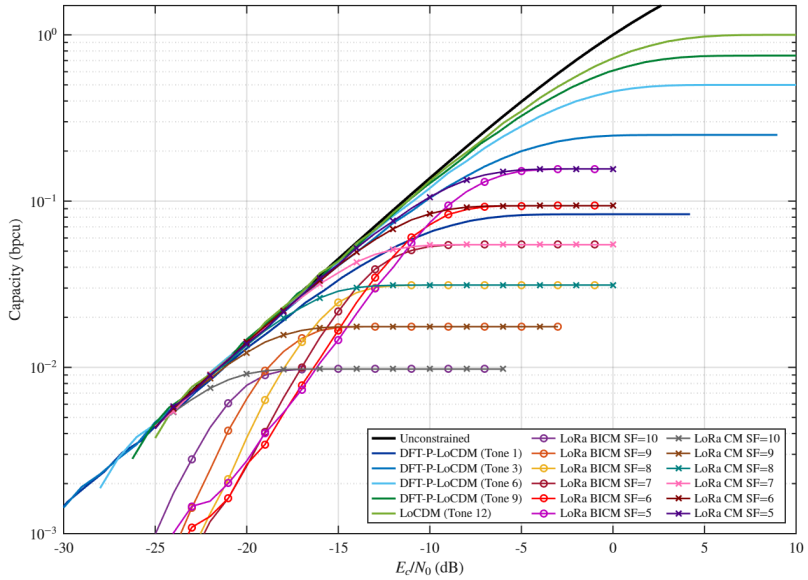
- In the uplink, DFT-P-LoCDM introduces a precoding step. A block of K time-domain symbols (s_m) first undergoes a K-point DFT:

$$S_l = \frac{1}{\sqrt{K}} \sum_{m=0}^{K-1} s_m e^{-j2\pi \frac{ml}{K}}, \quad l \in \{0, 1, 2, \cdots, K-1\}$$

CCDF of PAPR

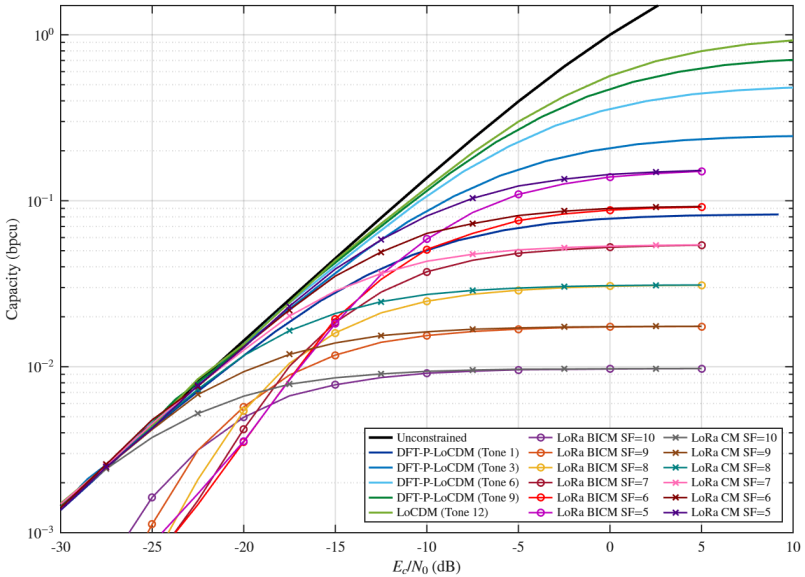


Capacity (AWGN Channel)

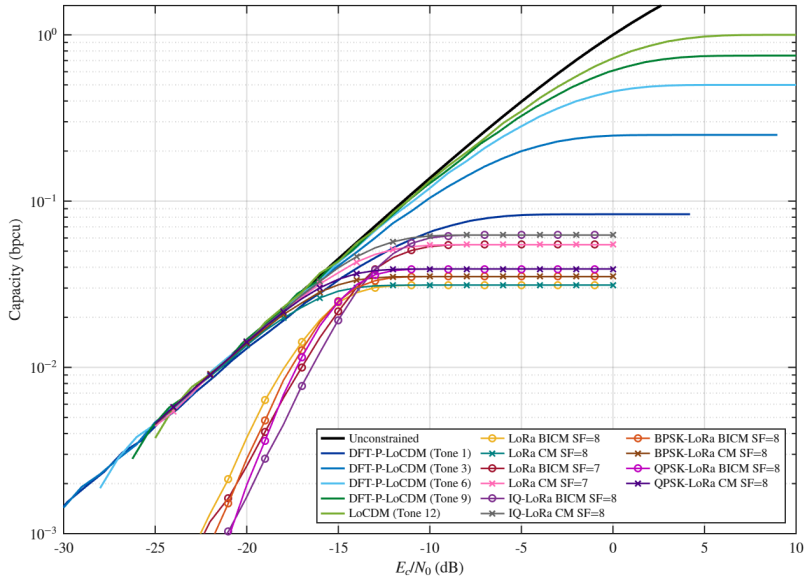


Performance

Capacity (Rayleigh Channel)



Comparison



- Improve achievable rated compared to legacy LoRa, and its extensions (PSK-LoRa, IQ-LoRa...).
- Achieve the spectral efficiency and data rates of NB-IoT.
- Interoperability enables LoCDM to use the same upper layer mechanisms and protocols as NB-IoT. But...
- The need to switch from random to scheduled radio access.

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Nonlinear Chirp Spread Spectrum

- Nonlinear chirps are among the solutions to increase network capacity.
- Studies have shown that it is possible to separate linear and nonlinear chirps (or two nonlinear chirps with different structures) at the receiver.
- The instantaneous frequency of a convex chirp:

$$f_{cvx}(t,k,s) = \frac{sB}{N} + B\left(\frac{t}{T}\right)^k - Bh\left(t - \tau_{s,cvx}^{(k)}\right)$$

with: $\tau_{s,cvx}^{(k)} = T\left(1 - \frac{s}{M}\right)^{\frac{1}{k}}$

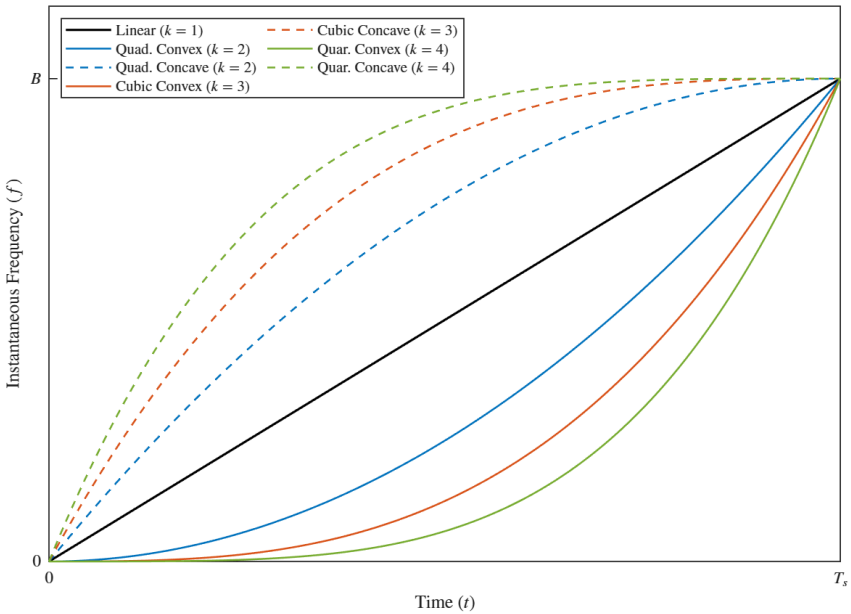
- The instantaneous frequency of a concave chirp:

$$f_{ccv}(t,k,s) = \frac{sB}{N} + B\left[1 - \left(1 - \frac{t}{T}\right)^k\right] - Bh\left(t - \tau_s^{(k)}\right)$$

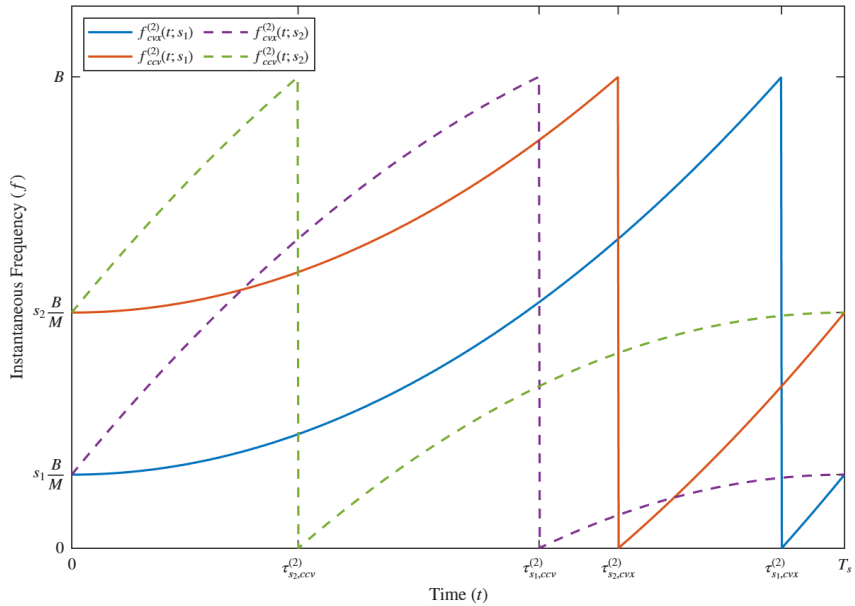
with: $\tau_{s,ccv}^{(k)} = T\left[1 - \left(\frac{s}{M}\right)^{\frac{1}{k}}\right]$

Nonlinear Chirp Spread Spectrum

Base chirp ($s = 0$).



Chirps for $s \neq 0$ and $k = 2$.



- An analysis of nonlinear chirps and their properties is still lacking in the literature.
- **Objective:** Conduct a comprehensive study of nonlinear chirp signals, evaluating among others: *Continuous Phase Modulation (CPM) properties, Cross-correlation, Receiver architecture, Spectral analysis...*

Thank you for your attention !



Questions?
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